Supplemental Material for 04/15 Book Reading Seminar

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Abstract

This is the supplemental material for the book reading seminar.
Please refer to the textbook and the whiteboard for the main content.
If necessary, please also refer to my repository.

About the textbook

an introduction to
Optimization on smooth manifolds

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I taught Riemannian optimization at EPFL in spring 2023 for the course MATH-512. The recorded lectures are available below. A semester runs for 14 weeks. Most weeks have a lecture and an exercise session, which each last 90-95 minutes (week 5 had double lectures). The students further learned the material by completing two extensive projects.

Video links bring you to an external hosting service. There, you can change the playback speed of videos (play faster, slower): see controls at the bottom-right of the video panel. You can also download the videos.

I gratefully acknowledge EPFL's CEDE (Matthew Goodman in particular) for video post-production!

#	Slides	Title + videos
Week 1		exercises
000	pdf	Logistics and prerequisites
001	pdf	Context and applications ▪ Context, 12 min ▪ Simple applications, 23 min (book §2.2, §2.4) ▪ More applications, 16 min (book §2.5, §2.6, §2.7)
002	pdf	Bird's-eye view and aims ▪ Video, 26 min (book §3.0)

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You may find the Manopt toolboxes helpful to use Riemannian optimization in your projects:

- Manopt in Matlab,
- PyManopt in Python,
- Manopt.jl in Julia.

For example, in Matlab, to minimize the function $f(x) = x^\top Ax$ on the sphere in $\mathbb{R}^n$, you can write:

```
n = 42;
A = randsym(n);
problem.M = spherefactory(n);
problem.cost = @(x) x'*A*x;
problem.egrad = @(x) 2*A*x;  % or use autodiff: problem = manoptAD(problem);
x = trustregions(problem);   % call an overly fancy algorithm
```

In Python, you could use JAX (among other backends) and write:

```
import pymanopt
import jax

n = 42
key = jax.random.key(0)
A = jax.random.normal(key, [n, n])
A = A + A.T

manifold = pymanopt.manifolds.Sphere(n)
@pymanopt.function.jax(manifold)
def cost(x):
    return x.T @ A @ x

problem = pymanopt.Problem(manifold, cost)
optimizer = pymanopt.optimizers.TrustRegions()
result = optimizer.run(problem)
```

There are lectures and software (Manopt) related to the textbook. They might be useful for understanding the textbook and doing exercises.

Lectures Manopt

Our goal

$$\min_{x \in \mathcal{M}} f(x) \tag{3.1}$$

Unit sphere

$$\mathrm{S}^{d-1} \coloneqq \{x \in \mathbb{R}^d : x^\top x = 1\}. \tag{3.2}$$

Personal Opinion: The sphere is actually quite a good example of a manifold, but I recommend NOT relying too much on it for intuition.

Smoothness

$$f \text{ is smooth} \iff f \text{ is of class } C^\infty.$$

Adapted from Wikipedia:

$$(U \subseteq \mathbb{R}^d \text{ is open}, \quad f\colon U \to \mathbb{R}, \quad k \in \mathbb{N})$$
$$f \text{ is of class } C^k \text{ on } U$$
$$\iff$$
$$\forall \alpha \in \mathbb{N}^d, |\alpha| \le k,\ \forall y \in U, \quad \frac{\partial^\alpha f}{\partial x_1^{\alpha_1} \partial x_2^{\alpha_2} \cdots \partial x_d^{\alpha_d}}(y_1, y_2, \ldots, y_d) \text{ exists and is continuous.}$$

It is known that smoothness is equivalent to the existence and continuity of the k-th Fréchet derivatives:

$$(U \subseteq \mathbb{R}^d \text{ is open}, \quad f\colon U \to \mathbb{R}, \quad k \in \mathbb{N})$$
$$f \text{ is of class } C^k \text{ on } U$$
$$\iff$$
$$\forall j \in \{1, 2, \ldots, k\}, \quad \forall y \in U, \quad \exists A\colon U \to \mathbb{R}^d, \quad \lim_{\|h\| \to 0} \frac{\|f(y+h) - f(y) - A(h)\|}{\|h\|} = 0.$$

Remark (partial differentiability vs. total differentiability): In general, partial differentiability does not imply total differentiability (i.e., Fréchet differentiability). However, the above statement holds. Why? This is because the smoothness requires not only existence but also continuity of the partial derivatives. The continuity of the partial derivatives ensures that the function behaves well enough to guarantee total differentiability.

Remark (smoothness implies continuity): If f is of class C^∞, then f is of class C^1, and thus f is continuous.

$$\begin{aligned}
\lim_{y \to x} f(y) &= f(x) + \lim_{y \to x} \mathrm{D}f(x)[y - x] + \lim_{y \to x} o(\|y - x\|) \\
&\le f(x) + \lim_{y \to x} \|\mathrm{D}f(x)\| \cdot \|y - x\| + \lim_{y \to x} o(\|y - x\|) \\
&= f(x).
\end{aligned}$$

Wiki: Smoothness Wiki: Fréchet derivative

Def 3.10

$$(\mathcal{E} \text{ is a } d\text{-dimensional linear space}, \quad \emptyset \neq \mathcal{M} \subseteq \mathcal{E})$$
$$\mathcal{M} \text{ is a smooth } \textit{embedded submanifold} \text{ of } \mathcal{E} \text{ of dimension } n$$
$$\iff$$

$$\Big(n = d \text{ and } \mathcal{M} \text{ is open in } \mathcal{E} \text{ (This } \mathcal{M} \text{ is } \textit{open submanifold}\text{)}\Big) \text{ or}$$
$$\Big(n = d - k \ (k \geq 1) \text{ and } \forall x \in \mathcal{M}, \quad \exists U_x \subseteq \mathcal{E} \text{ is open}, \quad \exists h_x \colon U_x \to \mathbb{R}^k \text{ is smooth},$$
$$(\forall y \in U_x, \quad h_x(y) = 0_k \iff y \in \mathcal{M}) \text{ and } (\operatorname{rank}(\mathrm{D}h_x(x)) = k)\Big).$$

The function h_x is called a *local defining function* of $\mathcal{M}$ at x.

Notation of U_x In this textbook, unlike in the general definition of a neighborhood, a neighborhood U of x is defined as an open set. To explicitly indicate that it is an open neighborhood of x, we use the notation "$\exists U_x$ is open" instead of simply writing "U."

Example: S^2 in $\mathbb{R}^3$

$$\mathrm{S}^2 = \{x \in \mathbb{R}^3 : x^\top x = 1\} = \{x \in \mathbb{R}^3 : h(x) = 0\}, \quad h(x) = x^\top x - 1.$$

We can check the conditions in Def 3.10:

$$n = 2, \quad k = 1, \quad d = 3,$$
$$h \colon U_x = \mathbb{R}^3 \subseteq \mathcal{E} = \mathbb{R}^3 \to \mathbb{R}, \quad h(x) = x^\top x - 1 \text{ is smooth},$$
$$h(y) = 0 \iff y^\top y = 1 \iff y \in S^2,$$
$$\operatorname{rank}(\mathrm{D}h(x)) = \operatorname{rank}(2x^\top) = 1 \quad (\text{since } x \neq 0).$$

Thus, S^2 is a smooth embedded submanifold of $\mathbb{R}^3$ of dimension $n = 2$.

Intuition of Def 3.10

$$\mathrm{D}h_x(x) \simeq \begin{pmatrix} \frac{\partial h_{x,1}}{\partial x_1} & \cdots & \frac{\partial h_{x,1}}{\partial x_d} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_{x,k}}{\partial x_1} & \cdots & \frac{\partial h_{x,k}}{\partial x_d} \end{pmatrix} \in \mathbb{R}^{k \times d},$$
$$\operatorname{rank}(\mathrm{D}h_x(x)) = k \iff \dim(\ker(\mathrm{D}h_x(x))) = d - k = n.$$

Roughly speaking, we have an n-dimensional local space along which h_x is constant and hence equal to 0, since $h_x(x) = 0$. By the definition of the local defining function h_x, this n-dimensional space is exactly $\mathcal{M} \cap U_x$. Thus, $\mathcal{M}$ is locally like an n-dimensional space, which motivates the tangent space $\mathrm{T}_x\mathcal{M}$ and the statement that the dimension of $\mathcal{M}$ is n.

Here is another intuition for the definition. It may seem that the cases $n = d$ and $n = d - k$ are defined differently, but they are actually the same.

$$n = d - 0 \text{ and } \forall x \in \mathcal{M}, \quad \exists U_x \subseteq \mathcal{E} \text{ is open}, \quad \exists h_x \colon U_x \to \mathbb{R}^0 = \{0\} \text{ is smooth(?)},$$
$$(\forall y \in U_x, \quad h_x(y) = 0 \iff y \in \mathcal{M} \iff U_x \subseteq \mathcal{M}) \text{ and } (\operatorname{rank}(\mathrm{D}h_x(x)) = 0).$$

Def 3.11 The textbook's definition is somewhat simpler than the general one:

$$(\mathcal{E}_1, \mathcal{E}_2 \text{ are linear spaces}, \quad U \subseteq \mathcal{E}_1, V \subseteq \mathcal{E}_2 \text{ are open})$$
$$F\colon U \to V \text{ is a diffeomorphism}$$
$$\Longleftrightarrow$$
$$F \text{ is a bijection and } F \text{ and } F^{-1} \text{ are smooth.}$$

Wiki: Diffeomorphism

Theorem 3.12

$$(\mathcal{E} \text{ is a } d\text{-dimensional linear space})$$
$$\mathcal{M} \subseteq \mathcal{E} \text{ is a } n = d - k\text{-dimensional smooth embedded submanifold of } \mathcal{E}$$
$$\Longleftrightarrow$$
$$\forall x \in \mathcal{M}, \quad \exists U_x \subseteq \mathcal{E} \text{ is open}, \quad \exists V \subseteq \mathbb{R}^d \text{ is open},$$
$$\exists F\colon U_x \to V \text{ is a diffeomorphism},$$
$$F(\mathcal{M} \cap U_x) = (\mathbb{R}^n \times \{0_k\}) \cap V.$$

Rewrite of Theorem 3.12

$$(\mathcal{E} \text{ is a } d\text{-dimensional linear space})$$
$$\mathcal{M} \subseteq \mathcal{E} \text{ is a } n = d - k\text{-dimensional smooth embedded submanifold of } \mathcal{E}$$
$$\Longleftrightarrow$$
$$\forall x \in \mathcal{M}, \quad \exists U_x \subseteq \mathcal{E} \text{ is open}, \quad \exists V \subseteq \mathbb{R}^d \text{ is open},$$
$$\exists F^{-1}\colon V \to U_x \text{ is a diffeomorphism},$$
$$F^{-1}((\mathbb{R}^n \times \{0_k\}) \cap V) = \mathcal{M} \cap U_x.$$

Non-degenerate $\mathrm{D}F^{-1}$

$$(\mathrm{D}F(x))^{-1} = \mathrm{D}F^{-1}(F(x)). \tag{3.20}$$

Since $\mathrm{D}F^{-1}(F(x)) \cdot \mathrm{D}F(x) = \mathrm{D}(F^{-1} \circ F)(x) = \mathrm{D}(\mathrm{id})(x) = I_d$, this is non-degenerate.

Def 3.14

$$(\mathcal{E} \text{ is a linear space}, \quad \mathcal{M} \subseteq \mathcal{E}, \quad x \in \mathcal{M})$$
$$\mathrm{T}_x\mathcal{M} \coloneqq \{c'(0) \mid \exists \varepsilon > 0, \quad \exists c\colon (-\varepsilon, \varepsilon) \to \mathcal{M} \text{ is smooth}, \quad c(0) = x\}. \tag{3.23}$$

Tangent space is a subset of the linear space

$$\mathrm{T}_x\mathcal{M} \subseteq \mathcal{E}.$$

This may look obvious, but it is not completely trivial.

$$c'(0) = \lim_{t \to 0} \frac{c(t) - c(0)}{t}.$$

Since $c(t) \in \mathcal{M} \subseteq \mathcal{E}$ and $c(0) = x \in \mathcal{M} \subseteq \mathcal{E}$, we have $c(t) - c(0) \in \mathcal{E}$ for all t. Therefore, the limit is also in $\mathcal{E}$. Hence, $\mathrm{T}_x\mathcal{M} \subseteq \mathcal{E}$.
Question: $c'(0) \in \mathcal{M}$? The answer is no. $\mathcal{M}$ is not closed under addition and scalar multiplication.

Theorem 3.15

$$\begin{gathered}(\mathcal{E} \text{ is a } d\text{-dimensional linear space}, \quad \mathcal{M} \subseteq \mathcal{E} \text{ is} \\ \text{a smooth embedded submanifold of dimension } n, \quad x \in \mathcal{M})\end{gathered}$$

$$\begin{cases} n = d & \implies \mathrm{T}_x\mathcal{M} = \mathcal{E}, \\ n = d - k \ (k \geq 1) & \implies \mathrm{T}_x\mathcal{M} = \ker(\mathrm{D}h_x(x)) \quad (\text{where } h_x \text{ is in Def 3.10}) \end{cases}$$

$\mathrm{T}_x\mathcal{M} \subseteq \mathcal{E}$ is already stated. We will show $\mathrm{T}_x\mathcal{M} \supseteq \mathcal{E}$.

Case $n = d$: It suffices to show that

$$\forall v \in \mathcal{E}, \quad \exists \epsilon > 0, \quad \exists c \colon (-\epsilon, \epsilon) \to \mathcal{M} \text{ is smooth}, \quad c(0) = x, \quad c'(0) = v.$$

Since $\mathcal{M}$ is open and $x \in \mathcal{M}$, we have

$$\exists \epsilon > 0, \quad \{x + tv : t \in (-\epsilon, \epsilon)\} \subseteq \mathcal{M}.$$

Then, $c(t) := x + tv$ satisfies the required conditions.

Case $n = d - k$:
Step1: $\mathrm{T}_x\mathcal{M} \subseteq \ker(\mathrm{D}h_x(x))$ Recall Definition 3.10.

$$\forall v \in \mathrm{T}_x\mathcal{M}, \quad \exists \epsilon > 0, \quad \exists c \colon (-\epsilon, \epsilon) \to \mathcal{M} \text{ is smooth}, \quad c(0) = x, \quad c'(0) = v. \forall c(t) \in U_x, \quad c(t) \in \mathcal{M} \implies$$

$$0_k = \frac{\mathrm{d}}{\mathrm{d}t} h_x(c(t)) = \mathrm{D}h_x(c(t))[c'(t)] \implies \mathrm{D}h_x(x)[v] = 0 \implies v \in \ker(\mathrm{D}h_x(x)).$$

Step2: $\mathrm{T}_x\mathcal{M} \supseteq (n\text{-dimensional subspace of } \mathcal{E})$. Recall Theorem 3.12.

$$\begin{gathered}\exists U_x \subseteq \mathcal{E} \text{ is open}, \quad \exists V \subseteq \mathbb{R}^d \text{ is open}, \\ \exists F^{-1} \colon V \to U_x \text{ is a diffeomorphism}, \\ F^{-1}((\mathbb{R}^n \times \{0_k\}) \cap V) = \mathcal{M} \cap U_x.\end{gathered}$$

We can construct a curve c to use the definition of $\mathrm{T}_x\mathcal{M}$ as follows:

$$\begin{gathered}\forall u \in \mathbb{R}^n, \quad \exists \epsilon > 0, \quad c \colon (-\epsilon, \epsilon) \to \mathbb{R}^d, \quad c(t) = F^{-1}(F(x) + t(u, 0_k)) \text{ is smooth}, \\ c(0) = F^{-1}(F(x)) = x, \quad c'(0) = \mathrm{D}F^{-1}(F(x))[(u, 0_k)].\end{gathered}$$

Thus, $\mathrm{T}_x\mathcal{M}$ contains the n-dimensional subspace, which is $\ker(\mathrm{D}h_x(x))$.

Definition 3.16

$$\dim\ \mathcal{M} := \dim \mathrm{T}_x\mathcal{M} \text{ at some } x \in \mathcal{M} \neq \emptyset.$$

This is exactly the n in Definition 3.10 (one of the most important points in this material).

The general definition of manifold

The textbook discusses the Definition 3.16 based on Theorem 3.15, so it might seem that the definition of dimension is only applicable to embedded submanifolds. However, as can be seen from the fact that the term "embedded submanifold" does not appear in Definition 3.16, the dimension can actually be defined for general manifolds, and rather the dimension itself is included in the definition of a manifold. In this textbook, we do not cover the precise definition of manifolds in order to get to optimization as quickly as possible. Therefore, as a supplement, we will introduce the general definition of manifolds and the definition of dimension for general manifolds.

Coordinate neighborhood:

$$(X \text{ is a topological space}, \quad U \subseteq X \text{ is open}, \quad U' \subseteq \mathbb{R}^m \text{ is open})$$
$$(U, \phi) \text{ is an } m\text{-dim. coordinate neighborhood} \iff \phi\colon U \to U' \text{ is a homeomorphism.}$$

Manifold:

$$(\mathcal{M} \text{ is a topological space})$$
$$\mathcal{M} \text{ is a smooth topological manifold of dimension } m$$
$$\iff$$
$$\begin{cases} (1) & \mathcal{M} \text{ is Hausdorff: } \forall x, y \in \mathcal{M}, x \neq y, \exists U_x, U_y \text{ are open}, U_x \cap U_y = \emptyset. \\ (2) & \exists \{(U_\alpha, \phi_\alpha)\}_{\alpha \in A} \ (m\text{-dimensional coordinate neighborhoods}), \quad M = \bigcup_{\alpha \in A} U_\alpha, \\ (3) & \forall \alpha, \beta \in A, U_\alpha \cap U_\beta \neq \emptyset, \quad \phi_\beta \circ \phi_\alpha^{-1}\colon \phi_\alpha(U_\alpha \cap U_\beta) \to \phi_\beta(U_\alpha \cap U_\beta) \text{ is smooth.} \end{cases}$$

Example 3.19

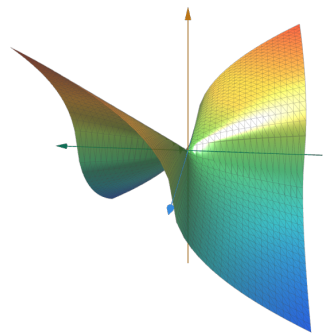

Figure 1: Visualization of $x_1 x_3 = x_2^2$ in $\mathbb{R}^3$. The orange axis represents x_2.

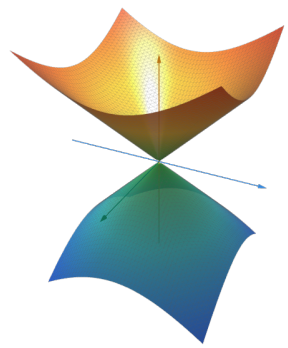

Figure 2: Visualization of $z_1^2 + z_2^2 = z_3^2$ in $\mathbb{R}^3$. The orange axis represents z_3.

We show the visualization in figs. 1 and 2.

Proposition 3.20 and Exercise 3.25

$$\Big(\mathcal{E}, \mathcal{E}' \text{ are } d, d'\text{-dimensional linear spaces,}$$
$$\mathcal{M} \subseteq \mathcal{E}, \mathcal{M}' \subseteq \mathcal{E}' \text{ are smooth embedded submanifolds of dimension } n, n'\Big)$$
$$\begin{cases} \mathcal{M} \times \mathcal{M}' \text{ is a smooth embedded submanifold of } \mathcal{E} \times \mathcal{E}', \\ \mathrm{T}_{(x,x')}(\mathcal{M} \times \mathcal{M}') = \mathrm{T}_x\mathcal{M} \times \mathrm{T}_{x'}\mathcal{M}'. \end{cases}$$

First part:
Case $n = d$ and $n' = d'$: $\mathcal{M} \times \mathcal{M}'$ is open in $\mathcal{E} \times \mathcal{E}'$ by the definition of the product topology.
Case $n = d$ and $n' = d' - k'$: omitted.
Case $n = d - k$ and $n' = d' - k'$:

$$\forall (x, x') \in \mathcal{M} \times \mathcal{M}', \quad \exists U_{x,x'} := U_x \times U_{x'} \subseteq \mathcal{E} \times \mathcal{E}' \text{ is open,}$$

$$\exists h_{x,x'} \colon U_{x,x'} \to \mathbb{R}^{k+k'} \text{ is smooth and defined by } h_{x,x'}(y,y') = (h_x(y), h_{x'}(y')),$$
$$\left(\forall (y,y') \in U_{x,x'}, \quad h_{x,x'}(y,y') = 0_{k+k'} \iff (y,y') \in \mathcal{M} \times \mathcal{M}' \text{ and}\right.$$
$$\left.\left(\operatorname{rank}(\mathrm{D}h_{x,x'}(x,x')) = \operatorname{rank}\left(\begin{pmatrix} \mathrm{D}h_x(x) & 0 \\ 0 & \mathrm{D}h_{x'}(x') \end{pmatrix}\right) = k + k'\right)\right).$$

Second part: Similar. Omitted.

Proposition 3.23 and Exercise 3.26 By definition. Omitted.

Lemma for Exercises 3.27–3.29

$$\mathcal{M} \in \{\mathcal{X}, \mathcal{C}, \mathcal{P}\}.$$

These exercises ask us to prove that the given manifolds are not embedded submanifolds of $\mathcal{E}$. We can check the conditions in Def 3.10. Note that $\mathcal{E} = \mathbb{R}^2$ with $d = 2$.
To solve these exercises, we first prove that if a manifold $\mathcal{M} \in \{\mathcal{X}, \mathcal{C}, \mathcal{P}\}$ is an embedded submanifold of $\mathbb{R}^2$, then the corresponding codimension k must be 1.

Case $n = d = 2$: Since $\mathcal{M}$ is not open in $\mathbb{R}^2$, $\mathcal{M}$ is not an open submanifold. Thus, $k \neq 0$.
Case $n = d - k$ ($k \geq 1$): Let h to be the local defining function. Since the dimension of $\mathrm{T}_x\mathcal{M}$ is at least 1 at some $x \in \mathcal{M}$ by the figures at p.26 of the textbook (and the definition), we have $\dim \ker \mathrm{D}h(x) \geq 1$. By the rank-nullity theorem, we have $\operatorname{rank}\mathrm{D}h(x) \leq 1$. Since $\operatorname{rank}\mathrm{D}h(x) = k$ and $k \geq 1$, it follows that $k = 1$.

Exercise 3.27

$$\mathcal{X} := \{x \in \mathbb{R}^2 : x_1^2 = x_2^2\}.$$

Assume that

$$\forall x \in \mathcal{X}, \quad \exists U_x \subseteq \mathbb{R}^2 \text{ is open}, \quad \exists h_x \colon U_x \to \mathbb{R}^k \text{ is smooth},$$
$$(\forall y \in U_x, \quad h_x(y) = 0_k \iff y \in \mathcal{X}) \text{ and } (\operatorname{rank}(\mathrm{D}h_x(x)) = k).$$

Consider lines $x_1 = x_2$ and $x_1 = -x_2$. We have $h_x(y) = 0$ for any point y on these lines. Thus, $\mathrm{D}h_x(x)\begin{bmatrix}1\\1\end{bmatrix} = 0$ and $\mathrm{D}h_x(x)\begin{bmatrix}1\\-1\end{bmatrix} = 0$. This means that $\mathrm{D}h_x(x)$ is degenerate. However, $\operatorname{rank}(\mathrm{D}h_x(x)) = k \geq 1$, which is a contradiction. Thus, $\mathcal{X}$ is not a smooth embedded submanifold of $\mathbb{R}^2$.

Exercise 3.28

$$\mathcal{C} := \{x \in \mathbb{R}^2 : x_1^2 = x_2^3\}.$$

Let us compute the tangent space at the origin $0 \in \mathcal{C}$:

$$\mathrm{T}_0\mathcal{C} = \{c'(0) \mid \exists \varepsilon > 0, \quad \exists c\colon (-\varepsilon, \varepsilon) \to \mathcal{C} \text{ is smooth}, \quad c(0) = 0\}.$$

On $\mathcal{C}$, we can only approach the origin along the positive x_2-axis. Thus, $c'(0) = 0$; otherwise, c cannot be smooth (this can be shown by contradiction). Thus, $\mathrm{T}_0\mathcal{C} = \{0\}$, which is a degenerate subspace of $\mathbb{R}^2$. However, $\mathrm{rank}(\mathrm{D}h_0(0)) = k = 1$, which is a contradiction. Thus, $\mathcal{C}$ is not a smooth embedded submanifold of $\mathbb{R}^2$.

Exercise 3.29

$$\mathcal{P} := \{x \in \mathbb{R}^2 : x_1^2 = x_2^4\}.$$

Apply Theorem 3.12 at the origin $0 \in \mathcal{P}$:

$$\begin{gathered}\exists U_0 \subseteq \mathbb{R}^2 \text{ is open}, \quad \exists V \subseteq \mathbb{R}^2 \text{ is open}, \\ \exists F^{-1}\colon V \to U_0 \text{ is a diffeomorphism}, \\ F^{-1}((\mathbb{R}^1 \times \{0_1\}) \cap V) = \mathcal{P} \cap U_0. \quad (\bigstar)\end{gathered}$$

Since $0 \in \mathcal{P} \cap U_0$ by definition, we have $F(0) \in (\mathbb{R}^1 \times \{0_1\}) \cap V$ by $(\bigstar)$. Let us define $F(0) = (a, 0)$ for some $a \in \mathbb{R}$. Since V is open, there exists $\epsilon > 0$ such that $(a + t, 0) \in V$ for all $t \in (-\epsilon, \epsilon)$. Let us define

$$c(t) = F^{-1}(a + t, 0) \quad (t \in (-\epsilon, \epsilon)).$$

Since F^{-1} is a diffeomorphism, c is also a diffeomorphism, i.e., bijective, invertible, and both c and c^{-1} are smooth. It is obvious that $\{(a+t, 0) : t \in (-\epsilon, \epsilon)\} \subseteq (\mathbb{R}^1 \times \{0_1\}) \cap V$. Thus, $\{c(t) : t \in (-\epsilon, \epsilon)\} \subseteq \mathcal{P} \cap U_0$ by $(\bigstar)$. Since c is continuous and $c(0) = F^{-1}(a, 0) = F^{-1}(F(0)) = 0$, its image contains points of $\mathcal{P}$ arbitrarily close to 0. Thus, there exist $\delta_1, \delta_2, \delta_3, \delta_4 > 0$ such that

$$\begin{gathered}\{(+x^2, +x) : x \in (0, \delta_1)\} \subseteq \{c(t) : t \in (-\epsilon, \epsilon)\}, \\ \{(-x^2, +x) : x \in (0, \delta_2)\} \subseteq \{c(t) : t \in (-\epsilon, \epsilon)\}, \\ \{(+x^2, -x) : x \in (0, \delta_3)\} \subseteq \{c(t) : t \in (-\epsilon, \epsilon)\}, \\ \{(-x^2, -x) : x \in (0, \delta_4)\} \subseteq \{c(t) : t \in (-\epsilon, \epsilon)\}.\end{gathered}$$

Thus, we have 4 distinct intervals as follows:

$$\begin{gathered}\{c^{-1}(F((+x^2, +x))) : x \in (0, \delta_1)\} \subseteq (-\epsilon, \epsilon), \\ \{c^{-1}(F((-x^2, +x))) : x \in (0, \delta_2)\} \subseteq (-\epsilon, \epsilon), \\ \{c^{-1}(F((+x^2, -x))) : x \in (0, \delta_3)\} \subseteq (-\epsilon, \epsilon), \\ \{c^{-1}(F((-x^2, -x))) : x \in (0, \delta_4)\} \subseteq (-\epsilon, \epsilon).\end{gathered}$$

Indeed, since c is invertible and smooth (i.e., continuous), these sets are intervals in $(-\epsilon, \epsilon)$. They are also distinct, since c^{-1} and F are bijective and the domains above are four distinct sets. We also have that a sequence in $\{c^{-1}(F((+x^2, +x))) : x \in (0, \delta_1)\}$ converges to 0 as $x \to 0$, and the same holds for the other three sets. Thus, we obtain four distinct intervals in $(-\epsilon, \epsilon)$ that approach 0. This is a contradiction, since there can be at most two such intervals in $(-\epsilon, \epsilon)$ that approach 0. Thus, $\mathcal{P}$ is not a smooth embedded submanifold of $\mathbb{R}^2$.

Definition 3.30

$$(\mathcal{M}, \mathcal{M}' \text{ are smooth embedded submanifolds of } \mathcal{E}, \mathcal{E}' \text{ respectively, } x \in \mathcal{M})$$
$$F \colon \mathcal{M} \to \mathcal{M}' \text{ is smooth at } x$$
$$\iff$$
$$(\exists U_x \subseteq \mathcal{E} \text{ is open}, \quad \exists \bar{F} \colon U_x \to \mathcal{E}' \text{ is smooth}, \quad \bar{F}(y) = F(y) \text{ for all } y \in \mathcal{M} \cap U_x).$$

Note that the smoothness of $\bar{F}$ is already defined at the beginning of this material. $\bar{F}$ is called a *local smooth extension* of F at x.

$$F \text{ is smooth} \iff \forall x \in \mathcal{M}, \quad F \text{ is smooth at } x.$$

Proposition 3.31

$$(\mathcal{M}, \mathcal{M}' \text{ are smooth embedded submanifolds of } \mathcal{E}, \mathcal{E}' \text{ respectively, } x \in \mathcal{M})$$
$$F \colon \mathcal{M} \to \mathcal{M}' \text{ is smooth}$$
$$\iff$$
$$\exists U \subseteq \mathcal{E} \text{ is open}, \quad \mathcal{M} \subseteq U \quad \exists \bar{F} \colon U \to \mathcal{E}' \text{ is smooth}, \quad \bar{F}(y) = F(y) \text{ for all } y \in \mathcal{M}.$$

Proof of Proposition 3.31

Only if part: Difficult. We use the paracompactness of $\mathcal{M}$ and the partition of unity.

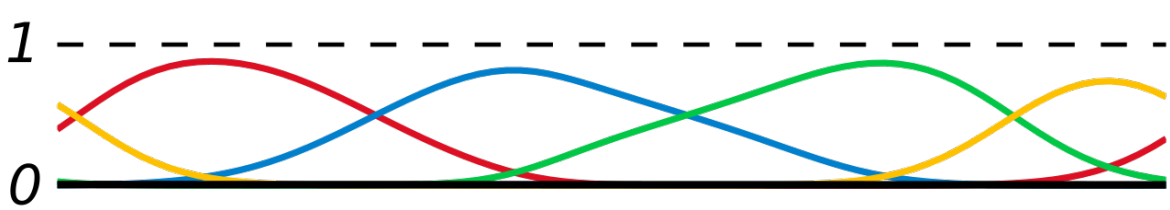

Figure 3: Cited from Wikipedia.

$$\exists \{U_{x_i}\}_{i \in I} \text{ is a finite open cover of } \mathcal{M},$$
$$\exists \{\phi_i\}_{i \in I} \text{ is a partition of unity subordinate to } \{U_{x_i}\}_{i \in I}.$$

Then, we can construct a local smooth extension $\bar{F}$ of F as follows:

$$\bar{F}(y) = \sum_{i \in I} \phi_i(y) \bar{F}_{x_i}(y) \quad \left(y \in U := \bigcup_{i \in I} U_{x_i} \right),$$

where $\bar{F}_{x_i}$ is a local smooth extension of F at x_i. We can check that $\bar{F}$ is a local smooth extension of F at any point $x \in \mathcal{M}$ as follows:

$$\bar{F}(y) = \sum_{i \in I} \phi_i(y) \bar{F}_{x_i}(y) = \sum_{i \in I} \phi_i(y) F(y) = F(y) \quad (y \in \mathcal{M}).$$

The smoothness of $\bar{F}$ is trivial since it is a finite sum of products of smooth functions.

If part: trivial.

Wiki: Paracompact space Wiki: Partition of unity

Exercise 3.32

$$\mathrm{Sym}(2)_1 \subseteq U = \mathrm{Sym}(2) \setminus \{0\} \subseteq \mathcal{E} = \mathrm{Sym}(2).$$

Let f be a smooth function on U which is defined on $\mathcal{E} = \mathrm{Sym}(2)$ as follows:

$$f(X) = \begin{cases} \frac{1}{\|X\|} & X \neq 0, \\ 0 & X = 0. \end{cases}$$

Then, a smooth extension of f from U to $\mathcal{E}$ does not exist, since it cannot be smooth at 0.

Exercise 3.37

We add the hypothesis that $\mathcal{M}$ is a smooth embedded submanifold of $\mathcal{E}$. Otherwise, the statement may be undefined in the context of this textbook. By Proposition 3.31, we have

$$\begin{aligned} &\exists U_1 \subseteq \mathcal{E} \text{ is open}, \ \mathcal{M} \subseteq U_1, \quad \exists \bar{F}_1 \colon U_1 \to \mathcal{E}' \text{ is smooth}, \quad \bar{F}_1(y) = F(y) \text{ for all } y \in \mathcal{M}, \\ &\exists U_2 \subseteq \mathcal{E} \text{ is open}, \ \mathcal{M} \subseteq U_2, \quad \exists \bar{F}_2 \colon U_2 \to \mathcal{E}' \text{ is smooth}, \quad \bar{F}_2(y) = F(y) \text{ for all } y \in \mathcal{M}. \end{aligned}$$

Thus, we have

$$\begin{gathered} \mathcal{M} \subseteq U_1 \cap U_2 \subseteq \mathcal{E}, \quad \bar{F} := a_1 \bar{F}_1 + a_2 \bar{F}_2 \colon U_1 \cap U_2 \to \mathcal{E}' \text{ is smooth}, \\ \bar{F}(y) = a_1 \bar{F}_1(y) + a_2 \bar{F}_2(y) = a_1 F_1(y) + a_2 F_2(y) = F(y) \quad (y \in \mathcal{M}). \end{gathered}$$

This means that F is smooth. By Proposition 3.35, we also have

$$\mathrm{D}F(x) = \mathrm{D}\bar{F}(x)|_{\mathrm{T}_x\mathcal{M}} = a_1 \mathrm{D}\bar{F}_1(x)|_{\mathrm{T}_x\mathcal{M}} + a_2 \mathrm{D}\bar{F}_2(x)|_{\mathrm{T}_x\mathcal{M}} = a_1 \mathrm{D}F_1(x) + a_2 \mathrm{D}F_2(x).$$

Exercise 3.38, 3.39

Almost the same as Exercise 3.37. Omitted.

Exercise 3.40

Almost the same as Exercise 3.37. We only show the case $\mathcal{M} = \mathbb{R}^{m_1}, \mathcal{M}' = \mathbb{R}^{m_2}, \mathcal{N} = \mathbb{R}^{n_1+n_2}$. Take an arbitrary point $(x, y) \in \mathcal{M} \times \mathcal{M}'$, and define

$$F_1^{(x,y)} \colon \mathbb{R}^{m_1} \to \mathbb{R}^{n_1}, F_1(x') = F(x', y), \quad F_2^{(x,y)} \colon \mathbb{R}^{m_2} \to \mathbb{R}^{n_2}, F_2(y') = F(x, y').$$

Then, we have

$$\mathrm{D}F(x,y)[(u,v)] = \begin{bmatrix} \mathrm{D}F_1^{(x,y)}(x) & 0 \\ 0 & \mathrm{D}F_2^{(x,y)}(y) \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \mathrm{D}F_1^{(x,y)}(x)u + \mathrm{D}F_2^{(x,y)}(y)v.$$

Exercise 3.41

Assumptions:

$$\mathcal{M} \subseteq \mathcal{E} \text{ is a } d\text{-dimensional smooth embedded submanifold,}$$
$$\ell \geq 1, \quad g \colon \mathcal{M} \to \mathbb{R}^\ell \text{ is smooth}, \quad \operatorname{rank}(\mathrm{D}g(x)) = \ell \text{ for all } x \in g^{-1}(0),$$
$$\mathcal{N} = g^{-1}(0) \subseteq \mathcal{M}.$$

Goal:

$$\mathcal{N} \text{ is a smooth embedded submanifold of } \mathcal{E},$$
$$\dim \mathcal{N} = \dim \mathcal{M} - \ell,$$
$$\mathrm{T}_x\mathcal{N} = \ker(\mathrm{D}g(x)) \subset \mathrm{T}_x\mathcal{M}.$$

By Definition 3.10, we can rewrite the assumptions and the goal as follows.

Assumptions(rw):

$$\forall x \in \mathcal{M}, \quad \exists U_x \subseteq \mathcal{E} \text{ is open}, \quad \exists h_x \colon U_x \to \mathbb{R}^{d-\dim \mathcal{M}} \text{ is smooth},$$
$$(\forall y \in U_x, \quad h_x(y) = 0_{d-\dim \mathcal{M}} \iff y \in \mathcal{M}) \text{ and } (\operatorname{rank}(\mathrm{D}h_x(x)) = d - \dim \mathcal{M}),$$
$$\ell \geq 1, \quad g \colon \mathcal{M} \to \mathbb{R}^\ell \text{ is smooth}, \quad \operatorname{rank}(\mathrm{D}g(x)) = \ell \text{ for all } x \in g^{-1}(0),$$
$$\mathcal{N} = g^{-1}(0) \subseteq \mathcal{M}.$$

Goal(rw):

$$\forall x \in \mathcal{N}, \quad \exists U_x \subseteq \mathcal{E} \text{ is open}, \quad \exists h_x \colon U_x \to \mathbb{R}^{d-\dim \mathcal{M}+\ell} \text{ is smooth},$$
$$(\forall y \in U_x, \quad h_x(y) = 0_{d-\dim \mathcal{M}+\ell} \iff y \in \mathcal{N}) \text{ and } (\operatorname{rank}(\mathrm{D}h_x(x)) = d - \dim \mathcal{M} + \ell).$$

Then, we can construct a local defining function for $\mathcal{N}$ at x as follows:

$$h'_x \colon U_x \to \mathbb{R}^{d-\dim \mathcal{M}+\ell}, \quad h'_x(y) = \begin{bmatrix} h_x(y) \\ g(y) \end{bmatrix},$$

$$\forall y \in U_x, \quad h'_x(y) = 0_{d-\dim \mathcal{M}+\ell} \iff h_x(y) = 0_{d-\dim \mathcal{M}} \text{ and } g(y) = 0_\ell \iff y \in \mathcal{N},$$
$$\operatorname{rank}(\mathrm{D}h'_x(x)) = \operatorname{rank}\left(\begin{bmatrix} \mathrm{D}h_x(x) \\ \mathrm{D}g(x) \end{bmatrix}\right) = d - \dim \mathcal{M} + \ell \text{ for all } x \in \mathcal{N}.$$

This h'_x is exactly the local defining function for $\mathcal{N}$ at x, completing the proof.